

LAB IA - 1

1. Write a Python program using Pandas to create a Series and a DataFrame with sample data, display them on the screen, select specific rows and columns using indexing, and rename a column and index values so that the difference between a Series and a DataFrame becomes clear.
2. Write a Python program that loads a dataset into a Pandas DataFrame, cleans the data by removing duplicates or handling missing values, groups the data based on a column, and then visualizes the grouped results using bar charts or line charts through Pandas plotting.
3. Write a Python program using Matplotlib that plots a line graph with markers, labels, and grid, and also creates a scatter plot, bar chart, and histogram using subplots in a single window to compare different graph types.
4. Given a dataset containing RegNo, FullName, Dept, JoinDate, Sales, Units, and Score, calculate the total experience in days for each record and classify the experience as “New” or “Experienced” based on a logical condition.
5. Build a summary section that shows total sales, average salary, and employee count for a selected department, where the department selection can be changed dynamically.

6. Create an interactive lookup panel where entering a registration number returns multiple related details, and display a custom message when the registration number does not exist.
7. Using a criteria range with more than one condition, compute the highest salary and average score for records meeting all the conditions.
8. Generate a dynamic list that removes duplicates, sorts results by a numeric column in descending order, and updates automatically when new data is added.